Experiment 1: Design Zero-shot, One-shot, and Few-shot Prompts to Generate a Product Description for a Smart Fitness Watch

Aim

To design and compare Zero-shot, One-shot, and Few-shot prompting techniques for generating an attractive and informative product description for a **Smart Fitness Watch** using a Large Language Model (LLM).

Procedure

1. Start.
2. Define the product as a Smart Fitness Watch.
3. Create a Zero-shot prompt without providing any examples.
4. Create a One-shot prompt with one example of a product description.
5. Create a Few-shot prompt with multiple examples of product descriptions.
6. Submit each prompt to the language model.
7. Generate product descriptions using all three prompting techniques.
8. Compare the outputs based on clarity, creativity, and relevance.
9. Display the generated descriptions.
10. Stop.

Result

The following are sample outputs that a language model could generate from each prompt.

A. Zero-shot output

Smart Fitness Watch — Track Your Progress

Stay active and connected with the Smart Fitness Watch. Monitor your daily steps, heart rate, sleep patterns, and workout performance with ease. Its stylish design and convenient features make it suitable for everyday wear, while its long-lasting battery helps you stay on track throughout your day. Achieve your fitness goals with a smarter approach to health and wellness.

B. One-shot output

Smart Fitness Watch — Your Everyday Fitness Partner

Take control of your fitness journey with a Smart Fitness Watch designed for an active lifestyle. Track your steps, monitor your heart rate, analyze your sleep, and stay informed about your daily activity. Featuring a modern design, comfortable fit, and convenient battery life, this watch combines style and functionality to support your everyday wellness goals.

C. Few-shot output

Smart Fitness Watch — Smarter Tracking, Better Living

Upgrade your lifestyle with a Smart Fitness Watch that keeps your goals within reach. Track daily steps, monitor heart rate, review sleep patterns, and measure workout progress with convenient fitness features. Designed with a sleek appearance and comfortable fit, it transitions effortlessly from workouts to everyday activities. With dependable battery performance and easy-to-access activity insights, it helps you stay motivated and make every move count.

Comparison of Results

Technique	Examples provided	Expected characteristics
Zero-shot	0	Direct, general-purpose description
One-shot	1	Follows the style and structure of one example
Few-shot	3	Can better reflect patterns and tone from multiple examples

Result: The three prompting techniques were designed and demonstrated for generating a Smart Fitness Watch product description. Few-shot prompting can offer more consistent stylistic guidance when the examples are relevant and well-written.

Experiment2 : Design Zero-shot, One-shot, and Few-shot Prompts to Generate a Blog on "Applications of Artificial Intelligence in Healthcare".

Aim

To design and compare Zero-shot, One-shot, and Few-shot prompting techniques for generating a 200-word blog on "Applications of Artificial Intelligence in Healthcare" using a Large Language Model (LLM).

Procedure

1. Start.
2. Define the topic: Applications of Artificial Intelligence in Healthcare.
3. Create a Zero-shot prompt without examples.
4. Create a One-shot prompt with one example of a blog.
5. Create a Few-shot prompt with multiple examples of short blogs.
6. Submit each prompt to the language model.
7. Generate a 200-word blog for each prompt.
8. Compare the generated blogs based on relevance, clarity, structure, and content.
9. Display the prompts and sample outputs.
10. Stop.

Sample Result

A. Zero-shot prompt output

Writing

Applications of Artificial Intelligence in Healthcare

Artificial Intelligence (AI) is transforming healthcare by improving medical services, supporting doctors, and helping patients receive better care. AI technologies analyze large amounts of medical data quickly and identify patterns that may be difficult for humans to detect.

One major application of AI is disease diagnosis. AI systems can assist healthcare professionals in identifying conditions such as cancer and heart disease at earlier stages. In medical imaging, AI helps analyze X-rays, CT scans, and MRI images to highlight possible abnormalities.

AI also supports drug discovery by helping researchers analyze chemical compounds and identify promising treatments. This can make parts of the research process faster and more efficient. In personalized medicine, AI analyzes patient information to help clinicians select treatments suited to individual needs.

Virtual health assistants and chatbots can answer basic health questions, provide appointment reminders, and help patients manage routine healthcare tasks. AI can also support remote patient monitoring by analyzing information from wearable devices and identifying changes that may require medical attention.

Despite these benefits, healthcare AI presents challenges, including data privacy, bias, security, and the need for reliable validation. AI should support, rather than replace, qualified healthcare professionals.

In conclusion, AI has significant potential to improve healthcare access, efficiency, and decision-making when used responsibly and under appropriate clinical supervision.

B. One-shot prompt output

Writing

Applications of Artificial Intelligence in Healthcare

Artificial Intelligence (AI) is becoming an important part of modern healthcare. By analyzing medical information and recognizing patterns, AI can help healthcare professionals make informed decisions and improve patient services.

One important application is disease diagnosis. AI tools can assist doctors in identifying conditions such as cancer and heart disease. AI-powered medical imaging systems can examine X-rays, CT scans, and MRI images to flag suspicious areas for further review.

AI also contributes to drug discovery by helping researchers analyze large datasets and identify potential medicines. In personalized treatment, AI can help clinicians consider a patient's medical history and other relevant information when planning care.

Another useful application is the virtual health assistant. Chatbots can answer general health questions, remind patients about appointments, and provide basic guidance on healthcare services. Wearable devices combined with AI can also help monitor activity and vital signs, supporting remote patient care.

However, AI must be used carefully. Patient privacy, data security, biased algorithms, and inaccurate predictions are important concerns. Medical professionals must evaluate AI-generated recommendations before using them in clinical decisions.

In conclusion, AI can make healthcare more efficient and support better patient outcomes. With responsible development, strong privacy protections, and human oversight, AI can become a valuable partner in delivering safe and effective healthcare.

C. Few-shot prompt output

Writing

Applications of Artificial Intelligence in Healthcare

Artificial Intelligence (AI) is reshaping healthcare by making medical analysis more efficient and supporting personalized patient care. Its ability to process large datasets helps healthcare professionals discover patterns and make better-informed decisions.

AI is widely used to assist disease diagnosis. For example, AI systems can analyze medical images and highlight possible signs of cancer or other abnormalities for specialists to review. These tools can support earlier detection, although final diagnosis requires appropriate clinical evaluation.

Another significant application is drug discovery. AI helps researchers examine biological data, study chemical compounds, and identify potential drug candidates. This may reduce time spent on certain stages of research. AI also contributes to personalized medicine by helping clinicians assess patient characteristics and compare possible treatment options.

Virtual assistants and healthcare chatbots can help patients schedule appointments, receive reminders, and find general health information. Meanwhile, AI-supported wearable devices can track activity and selected health measurements, potentially alerting users or clinicians to changes that deserve attention.

Despite its advantages, AI raises concerns about patient confidentiality, algorithmic bias, reliability, and unequal access to technology. Healthcare organizations must test these systems carefully and protect sensitive information.

In conclusion, AI offers promising applications in diagnosis, research, monitoring, and treatment planning. When combined with professional judgment, ethical safeguards, and continuous evaluation, it can support more accessible, efficient, and patient-centered healthcare.

Comparison of Results

Technique	Examples used	Characteristics
Zero-shot	0	Generates content from instructions alone
One-shot	1	Uses one example to guide style and structure
Few-shot	2 or more	Uses multiple examples to guide tone and organization

Result: All three prompting techniques were designed and demonstrated for the topic "Applications of Artificial Intelligence in Healthcare." The sample outputs illustrate how examples can guide the structure and style of generated content.

Experiment 3: Text Summarization Using Zero-shot, One-shot, and Few-shot Prompting

Aim

To design Zero-shot, One-shot, and Few-shot prompts to summarize an article into exactly 50 words and compare the generated summaries based on accuracy, completeness, and readability.

Procedure

1. Start.
2. Select an article for summarization.
3. Design a Zero-shot prompt without examples.
4. Design a One-shot prompt with one example.
5. Design a Few-shot prompt with multiple examples.
6. Generate a 50-word summary using each prompt.
7. Evaluate the summaries based on accuracy, completeness, and readability.
8. Compare the results in a table.
9. Display the best-performing technique.
10. Stop.

Article Used

Topic: Benefits of Artificial Intelligence

Artificial Intelligence (AI) is transforming modern society by enabling machines to perform tasks that traditionally require human intelligence. AI is widely used in healthcare, education, transportation, agriculture, and business. In healthcare, AI assists doctors in disease diagnosis and medical image analysis. In education, it supports personalized learning and automated assessment. In transportation, AI powers navigation systems and autonomous vehicles. Businesses use AI to analyze data, improve customer service, and automate repetitive tasks. Despite its advantages, AI raises concerns about privacy, job displacement, and algorithmic

bias. Therefore, responsible development and ethical use of AI are essential to ensure that its benefits reach society safely and fairly.

Prompts and Generated Summaries

A. Zero-shot prompting

Prompt:

Writing

Summarize the following article in exactly 50 words. Preserve the main ideas, ensure factual accuracy, and use clear, simple language. Do not add information that is not present in the article.

Article: [Insert the article here]

Generated summary (50 words):

Writing

Artificial Intelligence is transforming healthcare, education, transportation, agriculture, and business. It supports disease diagnosis, personalized learning, navigation, data analysis, and task automation. However, AI also creates concerns about privacy, job displacement, and algorithmic bias. Responsible development and ethical use are necessary to ensure its benefits reach society safely and fairly.

B. One-shot prompting

Prompt:

Writing

Example: Article: Solar energy provides clean electricity, reduces dependence on fossil fuels, and helps protect the environment. However, installation costs and weather dependence remain challenges.

Summary: Solar energy provides clean electricity and reduces fossil fuel dependence, but installation costs and weather conditions present challenges.

Now summarize the following article in exactly 50 words. Follow the example's concise style and preserve the main points.

Article: [Insert the article here]

Generated summary (50 words):

Writing

Artificial Intelligence improves services across healthcare, education, transportation, agriculture, and business. It helps diagnose diseases, personalize education, improve navigation, analyze data, and automate tasks. Nevertheless, concerns include privacy risks, job displacement, and algorithmic bias. Ethical practices and responsible development are essential for ensuring that AI benefits society safely and fairly.

C. Few-shot prompting

Prompt:

Writing

Example 1: Article: Online education enables flexible learning through digital platforms. It provides access to resources but may create challenges involving internet access and student engagement. Summary: Online education offers flexible learning and digital resources, but internet access and student engagement remain challenges.

Example 2: Article: Electric vehicles reduce tailpipe emissions and reliance on petrol. However, charging infrastructure, battery costs, and charging time can limit adoption. Summary: Electric vehicles reduce emissions and petrol dependence, but charging infrastructure, battery costs, and charging time can hinder adoption.

Now summarize the following article in exactly 50 words. Focus on the central idea, key applications, benefits, and challenges. Use clear language and avoid unsupported claims.

Article: [Insert the article here]

Generated summary (50 words):

Writing

Artificial Intelligence supports healthcare, education, transportation, agriculture, and business through diagnosis assistance, personalized learning, navigation, data analysis, and automation. Although these applications improve efficiency and services, AI raises concerns about privacy, job displacement, and algorithmic bias. Responsible development and ethical implementation are necessary to distribute its benefits safely and fairly across society.

Comparison of Generated Summaries

Criteria	Zero-shot	One-shot	Few-shot
Accuracy	High	High	High
Completeness	Covers the main points	Covers the main points	Covers the main points
Readability	Clear and direct	Concise and structured	Organized and consistent
Examples provided	None	One	Multiple
Style guidance	Based on instructions	Based on one example	Based on multiple examples

Note: These ratings are qualitative assessments of the sample summaries. In an actual experiment, accuracy and completeness should be checked against the original article, and readability should be evaluated using consistent criteria.

Result

Zero-shot, One-shot, and Few-shot prompts were designed to summarize the same article in 50 words. All three methods produced concise summaries covering the major applications, benefits, and challenges of Artificial Intelligence.

Experiment 4 : Professional Email Generation Using Zero-shot, One-shot, and Few-shot Prompting

Aim

To design Zero-shot, One-shot, and Few-shot prompts to generate a professional email requesting leave due to illness and compare the generated emails based on tone, grammar, formatting, and completeness.

Procedure

1. Zero-shot Prompt

1. Enter a prompt asking the AI to write a professional sick-leave email without examples.
2. Generate the email using the AI model.
3. Record the output.

2. One-shot Prompt

1. Provide one example of a professional leave-request email.
2. Ask the AI to write a similar email requesting leave due to illness.
3. Record the output.

3. Few-shot Prompt

1. Provide two or more examples of professional leave-request emails.
2. Ask the AI to generate a sick-leave email following the examples.
3. Record the output.

Output

A. Zero-shot Output

Email

Subject

Sick Leave Request

Dear Manager,

I am writing to inform you that I am unwell and unable to attend work today. I kindly request sick leave for one day to rest and recover.

I will keep you informed if I require additional leave. Thank you for your understanding and consideration.

Best regards,
[Your Name]

B. One-shot Output

Email

Subject

Request for Sick Leave

Dear Manager,

I hope you are doing well. I am feeling unwell and would like to request sick leave for one day to recover from my illness.

I apologize for any inconvenience caused and appreciate your understanding. I will resume work as soon as I am feeling better.

Thank you for your consideration.

Regards,
[Your Name]

C. Few-shot Output

Email

Subject

Request for Medical Leave Due to Illness

Dear Sir/Madam,

I am writing to request medical leave as I am currently experiencing health issues and need time to rest and recover. I kindly request leave for the next two days, subject to my recovery.

I will ensure that any urgent pending tasks are communicated appropriately and will keep you updated regarding my return to work.

Thank you for your understanding and support.

Yours sincerely,
[Your Name]

Comparison of Generated Emails

Criteria	**Zero-shot**	**One-shot**	**Few-shot**
Tone	Professional and polite	Courteous and formal	Formal and detailed
Grammar	Correct	Correct	Correct
Formatting	Clear structure	Well-organized	Detailed structure
Completeness	Covers basic leave request	Includes request and courtesy	Includes request, duration, and work communication
Examples used	None	One	Multiple

6. Result

Zero-shot, One-shot, and Few-shot prompts were designed and used to generate professional emails requesting leave due to illness. The generated emails were compared based on tone, grammar, formatting, and completeness.

Experiment 5: Generate a Promotional Social Media Post for an AI Workshop Using Zero-shot, One-shot, and Few-shot Prompts

Aim

To design Zero-shot, One-shot, and Few-shot prompts to generate a promotional social media post for an AI Workshop and compare the outputs based on creativity, clarity, engagement, and promotional effectiveness.

Procedure

1. Zero-shot Prompt

1. Enter a prompt asking the AI to create a promotional social media post for an AI Workshop without providing examples.
2. Generate the post with an attractive headline, workshop highlights, and a call to action.
3. Record the generated output.

2. One-shot Prompt

1. Provide one example of a promotional social media post for a technology workshop.
2. Ask the AI to create a similar post promoting an AI Workshop.
3. Record the generated output.

3. Few-shot Prompt

1. Provide two or more examples of promotional posts for workshops or technology events.
2. Ask the AI to generate a promotional post for an AI Workshop using the examples as guidance.
3. Record the output and compare it with the other two posts.

Output

A. Zero-shot Output

Writing

Unlock the Future with Artificial Intelligence!

Ready to explore the exciting world of AI?

Join our **AI Workshop** and discover how Artificial Intelligence is transforming industries and solving real-world problems.

Learn the fundamentals of AI
Explore practical applications
Discover innovative AI tools and techniques
Understand how AI shapes the future

Whether you're a student, beginner, or technology enthusiast, this workshop is a great opportunity to expand your knowledge!

Register today and start your AI journey!

#AIWorkshop #ArtificialIntelligence #Innovation #FutureOfTechnology #LearnAI

B. One-shot Output

Writing

Step into the World of Artificial Intelligence!

Want to learn how AI works and how it can solve real-world challenges?

Join our **AI Workshop** to explore AI concepts, discover useful tools, and gain practical insights into emerging technologies.

Explore AI fundamentals
Discover real-world applications
Build your technology skills

Don't miss this opportunity to learn something new and prepare for the future!

Register now and be part of the AI revolution!

#AIWorkshop #LearnAI #Technology #Innovation #DigitalSkills

C. Few-shot Output

Writing

Discover AI. Create the Future!

Curious about Artificial Intelligence? Turn your curiosity into knowledge at our **AI Workshop!**

What you'll explore:

- Fundamental AI concepts and technologies
- Practical AI tools and hands-on learning opportunities
- Real-world applications of AI across different industries
- Emerging opportunities in the world of intelligent technology

Who should join? Students, beginners, aspiring developers, and technology enthusiasts eager to explore AI.

Learn new skills, spark innovative ideas, and take your first step toward understanding the technology shaping tomorrow.

Don't wait—register today and start exploring AI!

#AIWorkshop #ArtificialIntelligence #FutureReady #TechInnovation #LearnByDoing

Comparison of Generated Outputs

Criteria	**Zero-shot**	**One-shot**	**Few-shot**
Creativity	Attractive	Follows the example's style	Varied and engaging
Clarity	Clear and informative	Simple and direct	Well-structured
Engagement	Uses emojis and hashtags	Uses a concise call to action	Uses sections and audience targeting
Promotional effectiveness	Highlights benefits	Follows a familiar promotional format	Provides detailed workshop highlights
Examples used	None	One	Multiple

Result

Zero-shot, One-shot, and Few-shot prompts were designed to generate promotional social media posts for an AI Workshop. The generated posts were compared based on creativity, clarity, engagement, and promotional effectiveness.